

## Module 5: Membranes - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Why do phospholipids arrange themselves into a bilayer in water?
2. A. Their heads avoid contact with water.
3. B. They are completely hydrophobic.
4. C. Water-loving heads face outward while water-fearing tails tuck inward.
5. D. They dissolve completely in water.
6. Answer: C
  
7. Why: The hydrophilic heads face the water and the hydrophobic tails hide inside, forming a stable two-layer membrane.
  
8. A small nonpolar molecule such as oxygen gas crosses the membrane mainly by:
9. A. simple diffusion down its concentration gradient
10. B. active transport that spends ATP
11. C. not crossing at all
12. D. osmosis of water
13. Answer: A
  
14. Why: Small nonpolar molecules pass directly through the bilayer from high to low concentration, with no energy required.
  
15. An animal cell is placed in pure water, a hypotonic solution. What happens?
16. A. Water leaves and the cell shrinks.
17. B. Nothing moves in or out.
18. C. Salt rushes in and the cell shrinks.
19. D. Water enters and the cell may swell or burst.
20. Answer: D
  
21. Why: In a hypotonic solution water moves into the cell by osmosis, so the cell swells and can burst.

22. Moving ions against their concentration gradient requires:

23. A. only ordinary diffusion

24. B. active transport that uses energy

25. C. no membrane proteins

26. D. removal of the membrane

27. Answer: B

28. Why: Movement against a gradient needs energy (ATP) and transport proteins, which is active transport.